LEARNING STYLE DETECTION AND DROPOUT PREDICTION SYSTEM

Project Group Details

Sl. No.	Roll Number	Name of Student
1	B231099CS	MUHAMMAD SWALIH K M
2	B230617CS	VIVEKANAND B
3	B230058CS	MOHAMMED HANIF

Project Supervisor(s): Dr. Amit Praseed
Dr.Anand Babu. N. B

21/1/2026.

Abstract

The project addresses the problem of high student dropout rates in higher education and the lack of personalization in standard learning platforms. Recent studies in this area indicate that aligning content with a student's learning style significantly improves engagement and performance, yet most platforms deliver identical content to all learners. Several approaches have been proposed, including explicit questionnaires like the Index of Learning Styles (ILS) and supervised machine learning classifiers, which have shown promising results under certain conditions.

However, the literature reveals several open problems such as the unreliability of self-reported surveys and the scarcity of labeled data required to train accurate detection models. Additionally, traditional dropout predictors often rely on grades alone, ignoring the critical behavioral patterns that indicate a mismatch in learning styles. These issues highlight the need for further investigation and improvement in automated, non-intrusive identification methods.

The primary objectives of this project are to develop a machine learning framework that automatically detects learning styles (based on the Felder-Silverman model) and utilizes this to predict student dropout probability. To achieve these objectives, the project plans to follow a structured methodology that includes extracting user interaction logs from the learning platform, applying a semi-supervised self-training algorithm to label learning styles from limited data, and training a predictive classifier. The outcome of this work is expected to contribute towards an adaptive early-warning system that facilitates personalized interventions to improve student retention.

Keywords: Student Dropout Prediction, FSLSM, Educational Data Mining, Learning Platform, Learning Analytics.